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SUPERSEDING
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MILITARY SPECIFICATION

CLEANER, STEAM, PRESSURE JET, TRAILER-MOUNTED (FOR BOMB DISPOSAL OPERATIONS)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers trailer-mounted, oil-fired, 180 gallons per hour (gph), pressure cleaners equipped with nonsparking cleaning guns. The cleaners shall be powered by a gasoline engine-driven generator.

1.2 Classification. Cleaners shall be of the following styles, as specified (see 6.2.1):

Style 1 - Two-wheel, vehicular-towed.
Style 2 - Four-wheel, vehicular-towed.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-C-437 - Cleaning Compound, High Pressure (Steam) Cleaner.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Measurement of Electromagnetic Interference Characteristics.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A-106 - Seamless Carbon Steel Pipe for High-Temperature Service.
D-396 - Fuel Oils.
D-975 - Diesel Fuel Oil.
D-1655 - Aviation Turbine Fuels.
D-3699 - Kerosine.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The cleaner shall consist essentially of a gasoline engine, generator, combustion chamber, pressure atomizing burner, blower, fuel and water tanks, pumps, and operational and safety controls. It shall be completely assembled on a wheel-mounted chassis to form a totally integrated, weatherproof, self-contained unit.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish one cleaner for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard commercial product. The cleaner shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the cleaner being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated

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in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5.1 Nonsparking materials. Metallic parts of the cleaning gun assembly shall be made of a nonsparking material. Acceptable nonsparking materials shall be a copper alloy or aluminum alloy. A certificate of compliance (see 6.2.2) shall be provided as evidence that nonsparking materials have been provided.

3.6 Operational requirements. The cleaner shall be of the steam cleaning, high-pressure/hot-wash, or combination steam and high-pressure/hot-wash type, as specified (see 6.2.1). All types of cleaners shall have sufficient capacity to generate the rated discharge pressures and temperatures within 5 minutes from a cold 60° Fahrenheit (F) start. The thermal efficiency, based on the relation of cleaner output to fuel input shall be not less than 75 percent when the cleaner is operated at its rated capacity. The burner shall be capable of automatic operation by remote control from the nozzle shutoff valve on the cleaner gun.

3.6.1 Steam cleaner. When a steam cleaner is specified (see 3.6), it shall be designed with sufficient capacity to heat not less than 180 gph of 60°F water to within 10°F of the saturation temperature for the rated operating pressure of the cleaner. The rated operating pressure shall be not less than 90 pounds-force per square inch gage (psig).

3.6.2 High-pressure/hot-wash cleaner. When a high-pressure/hot-wash cleaner is specified (see 3.6), it shall be designed with sufficient capacity to heat not less than 180 gph of 60°F water to not less than 200°F for the rated operating pressure. The rated operating pressure shall be not less than 650 psig.

3.6.3 Combination cleaner. When a combination steam and high-pressure/hot-wash cleaner is specified it shall meet the operational requirements of 3.6, 3.6.1, and 3.6.2.

3.7 Design and construction. The unit shall be designed to withstand the strains, jars, and vibrations incident to shipping, storage, and the conditions of mobile operation specified herein. The installation and location of components shall be such as to permit normal servicing and maintenance of the cleaner. The cleaner shall be constructed in a manner to permit operation in the open under adverse weather conditions. Individual components, including electrical controls and transformers, shall either be weatherproof or protected by weatherproof enclosures. When specified (see 6.2.1), a housing shall fully enclose the sides, top, and ends of the cleaner. The housing shall not be required for areas where components such as the combustion chamber or water or fuel tanks provide a suitable enclosure. When the housing is specified, doors or removable panels shall provide access to all interior components of the cleaner.

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3.8 Gasoline engine. The engine shall be a standard 4-cycle, air-cooled engine. The continuous horsepower output rating shall be sufficient to meet the input requirements of the generator, blower, and pumps without exceeding the manufacturer's published continuous horsepower rating for the speed at which the engine will normally operate. The engine shall be furnished complete with all accessories normally supplied as standard equipment, including a governor, air cleaner, and muffler.

3.9 Generator. A generator shall be provided as an integral part of the cleaner to furnish electrical power for operation of the cleaner ignition system and all electrical components. The generator shall be driven by the gasoline engine and shall have a voltage and amperage rating consistent with the combined requirements of the electrical components of the cleaner.

3.10 Heating coil. The heating coil shall be fabricated of steel tubing conforming to ASTM A106, grade A. After fabrication, the complete heating coil shall be capable of withstanding a hydrostatic test pressure of 1,000 psig when tested in accordance with 4.5.2. The heating coil assembly shall be designed and installed in a manner to permit its removal and replacement.

3.11 Combustion chamber. The combustion chamber shall be enclosed in an insulated sheet-metal jacket or casing. The products of combustion shall discharge from the top of the cleaner.

3.12 Burner. The burner shall be of the pressure-atomizing type and shall be capable of firing any of the following fuels:

a. Fuel oil conforming to ASTM D396, grade 1 or 2.
b. Diesel oil conforming to ASTM D975, grade 1-D or 2-D.
c. Kerosene conforming to ASTM D3699, grade 1-K.
d. Aviation turbine fuel conforming to ASTM D1655, type Jet A or Jet A-1.

The burner shall be furnished complete with continuous, high-voltage, spark ignition; blower fan; controls specified in 3.14; and all piping, fittings, and accessories necessary to form a complete and integrated burner assembly.

3.13 System requirements.

3.13.1 Water circulating system. The water circulating system shall include a water reservoir tank with a float-controlled inlet valve, a positive displacement water pump, a blowdown valve, and a pump pressure relief device. The float-controlled water inlet valve shall be capable of operating within a range of supply pressures up to 125 psig. A manually operated shutoff valve with a 0.75-inch hose adapter shall be installed ahead of the inlet on the float valve.

3.13.2 Fuel system. The burner fuel system shall include a fuel tank, filter, fuel pump, manual control valve, solenoid shutoff valve, and fuel flow controls specified in 3.14. The fuel tank shall have sufficient capacity for not less than 4 hours continuous operation of the cleaner.

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3.13.3 Electrical system. Wiring shall conform to and be installed in accordance with the applicable requirements of NFPA No. 70. Wire shall be moisture-, heat-, and oil-resistant. Control boxes and panels shall be furnished with drip-proof enclosures or shall otherwise be protected against exposure to moisture and weather in accordance with 3.7.

3.13.4 Concentrated cleaning solution system. Unless otherwise specified (see 6.2.1), a concentrated cleaning solution system shall be furnished as an integral part of the cleaner. The system shall be suitable for handling concentrated solutions prepared from a steam cleaning compound conforming to P-C-437. The system shall include a compound mixing tank and a means for introducing the concentrated cleaning solution into the incoming water supply or discharge piping. A device shall be provided for metering the feed rate of the concentrated cleaning solution. The metering device shall be adjustable and shall have a range of zero to no less than 12 gph of prepared solution. The tank for mixing and storing the concentrated solution shall have a capacity of not less than 15 gallons. A measure for apportioning cleaning compound during preparation of the solution shall be furnished.

3.14 Controls. The cleaner shall be equipped with all mechanical and electrical controls and devices necessary for safe and efficient operation, including but not limited to the following:

a. Safety control to shut down the burner in the event the temperature of the heating coil exceeds a predetermined safe limit due to partial or total interruption of the water supply to the coil. This control may be responsive to coil temperature, coil expansion, or water supply flow rates and storage levels at the option of the manufacturer.
b. Pressure limit device to prevent the pressure in the heating coil and delivery hose from exceeding a predetermined safe limit.
c. An automatic and manual fuel shutdown device to prevent flow of fuel to the burner in the event of ignition interruption or insufficient fuel pressure to insure proper atomization.
d. A control to automatically maintain the operating pressure in the heating coil to within ±10 percent of the specified operating pressure under varying rates of discharge from the cleaning gun.
e. A safety-relief valve to relieve excess water and steam pressure.

3.15 Components and accessories.

3.15.1 Gages. The following gages shall be provided and mounted in readily visible locations:

a. Heating coil inlet pressure gage.
b. Heating coil outlet pressure gage.
c. Burner fuel pressure gage.

Each gage shall have a scale range such that the normal operating pressure will register within 35 to 65 percent of the full-scale reading. Pulsation dampeners shall be included in the gage connection.

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3.15.2 Drain valves. Means shall be provided for complete draining of all systems of the cleaner. The drain connections shall include such pipe and fittings as may be required to provide ready accessibility for manual operation of the drain valves. Drain outlets shall be so located that drainage does not impinge on components of the cleaner.

3.15.3 Cleaning gun assembly. All metal parts of the cleaning gun, from the delivery hose to the end of the cleaning gun, including the nozzle, shall be fabricated of nonferrous, nonsparking materials specified in 3.5.1. Handles and heat guards may be fabricated of suitable plastic or other non-metallic materials. The cleaner shall be equipped with a hanger, rack, or other appurtenance for stowing the gun and hose when in transit. The cleaning gun assembly shall be furnished with the following:

a. 1 - round nozzle.
b. 1 - 2-inch flat nozzle.
c. 1 - 4-inch flat nozzle.
d. 2 - 50-foot lengths of 0.5-inch inside diameter delivery hose.

3.15.4 Supply hose. One 50-foot length of 0.75-inch water supply hose shall be furnished with each cleaner. The ends shall terminate in standard hose connections, one female and the other suitable for connection to the water inlet valve hose adapter. The hose shall be yarn-reinforced, natural or synthetic rubber.

3.15.5 Grounding strap. A grounding strap capable of preventing the accumulation of static charges shall be provided on the rear of the cleaner. The strap shall be copper and of sufficient length to insure positive contact with the ground.

3.15.6 Fire extinguisher. When specified (see 6.2.1), a fire extinguisher shall be furnished and mounted in a convenient location on the exterior of the cleaner. The extinguisher shall be of the dry chemical type, suitable for electrical or flammable liquid fires, and shall have a capacity of not less than 2.5 pounds.

3.16 Mounting. The cleaner shall be mounted on a trailer-type under-carriage of the style specified (see 1.2). The tires shall be pneumatic, not less than a four-ply rating, and in accordance with the applicable requirements of the T&RA Yearbook. The wheels shall be removable and rotate on antifriction ball or roller bearings. The trailer frame shall be supported on leaf springs designed for the intended application and conforming to SAE J510. The tow bar shall be equipped with a lunette eye and safety chain. The lunette eye shall be fabricated from 1-inch round stock with an inside diameter of 3 inches $\pm$0.0625. The safety chain shall be anchored to the draw bar and will terminate in a safety hook or pin-type clevis. Reflectors shall be mounted on the rear corners of the unit. The unit shall be strong, durable, and capable of withstanding the conditions of being towed at speeds up to 15 miles per hour (mph) over improved and unimproved roads (see 4.4.4 and 6.4).

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3.16.1 Style 1. The style 1 mountings shall have two wheels and a single axle. The trailer shall be equipped with a retractable and adjustable landing leg with caster. The landing leg shall be designed to support the trailer in a level position when uncoupled from the towing vehicle. A positive means shall be provided to secure the leg in the retracted position.

3.16.2 Style 2. Style 2 mountings shall have four wheels and two tandem axles. The trailer shall be equipped with a retractable and adjustable landing leg with caster. The landing leg shall be designed to support the trailer in a level position when uncoupled from the towing vehicle. A positive means shall be provided to secure the leg in the retracted position.

3.17 Lubrication. Unless otherwise specified (see 6.2.1), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pounds per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.18 Electromagnetic interference control. When specified (see 6.2.1), the cleaner shall conform to the electromagnetic interference suppression requirements of MIL-STD-461, class C1.

3.19 Lifting and tiedown attachments. When specified (see 6.2.1), the cleaner shall be equipped with lifting and tiedown attachments conforming to MIL-STD-209, class 3, type III.

3.20 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.21 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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3.22 Servicing and restoration. Each unit tested shall be serviced and restored to a condition equal to the original condition of the unit, excluding nominal wear incurred during the tests. The restoration shall include paint touchup or repainting, as required for delivery.

3.23 Special marking. The following notation shall be stenciled in a conspicuous location on the cleaner: USE KEROSENE, DIESEL OIL, LIGHT FUEL OIL, OR JP-5 FOR BURNER.

3.24 Identification plate. An identification plate will be furnished by the contracting officer for each cleaner. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each cleaner in a conspicuous place with brass screws or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.25 Workmanship. All equipment being furnished shall be made and finished with a degree of uniformity and a grade of workmanship practicable in a well-equipped factory.

3.25.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.25.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.25.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.25.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements speci-

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fied herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one cleaner when a first article is required (see 3.2 and 6.3). This inspection shall include the examination of 4.3 and the tests of 4.4 and 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on each cleaner.

4.3 Examination. Each cleaner shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. These tests shall be performed when a first article is specified (see 6.2.1). Failure of the cleaner to comply with any of the requirements specified herein shall constitute cause for rejection.

4.4.1 Performance test. The first article shall be tested to verify compliance with the requirements of 3.6. The test shall be conducted with use of one of the fuels specified in 3.12. The duration of the test shall be not less than 4 hours. Combination steam and high-pressure/hot-wash cleaners shall be tested for not less than 2 hours on each cycle. The following data shall be recorded at the end of the first 5 minutes of operation and at 30-minute intervals thereafter during the test:

a. Date.
b. Time.
c. Ambient temperature.

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d. Type of fuel and British thermal unit heat content per pound.
e. Temperature of incoming water supply.
f. Weight of water supplied to the cleaner.
g. Water temperature at heater outlet.
h. Pressure at heater outlet.
i. Weight of fuel burned.
j. Calculated thermal efficiency.
k. Calculated rate of discharge in gph.

The solution feed system shall be checked during the test for compliance with 3.13.4.

4.4.2 Endurance test. When specified (see 6.2.1), the first article shall be subjected to an endurance test of not less than 8 hours. The test shall be conducted with the cleaner operating at the maximum rated pressure and flow rate. Combination steam and high-pressure/hot-wash cleaners shall be tested for no less than 4 hours on each cycle. Failure of the cleaner to operate or the need for major repairs or replacement of parts shall be cause for rejection.

4.4.3 Safety control tests. After the test of 4.4.1, the first article shall be tested to verify compliance with 3.14. The conditions which activate the safety controls shall be simulated by interruption of water and fuel flows, as applicable, and for the pressure control test, by throttling the discharge.

4.4.4 Road test. Nozzles, hoses, and any other accessories shall be stowed and the cleaner shall be coupled to a suitable towing vehicle. The cleaner shall be towed a distance of not less than 10 miles over improved and unimproved roads (see 6.4), 5 miles over each or a combination of each, at speeds up to and including 15 mph. The unit shall satisfactorily meet the requirements of 3.16. After completion of the road test, the cleaner shall be tested in accordance with 4.5.1.

4.5 Quality conformance tests.

4.5.1 Functional test. Each cleaner shall be operated not less than 1 hour on a steam cleaning cycle and 1 hour on a high-pressure/hot-wash cycle, as applicable, to verify the functional adequacy of operational components and controls. Any cleaner requiring frequent adjustment or which suffers flame or ignition failure or fails to maintain the specified operational pressure, shall be rejected.

4.5.2 Hydrostatic test. Each heating coil after assembly and prior to installation shall be hydrostatically tested at a pressure of not less than 1,000 pounds per square inch for a period of not less than 1 minute. Any coil exhibiting leakage or rupture shall be rejected.

4.5.3 Electromagnetic interference suppression tests. To determine conformance to 3.18, the cleaner shall be subjected to the tests in accordance with MIL-STD-462. When suppressed to conform to 3.18, the manufacturer upon approval of the contracting officer may furnish a certification (see 6.2.2) in lieu of the test that the cleaner meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

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4.5.4 Lifting and tiedown attachment test. When the lifting and tiedown attachments are required (see 6.2.1), they shall be tested to determine conformance to 3.19.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.4.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.4.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.5 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type II, method IIa.

5.1.1.6 Burner and fuel tank. Interior surfaces of the burner and fuel tank shall be preserved with P-10, grade 30 preservative.

5.1.1.7 Water pump, valves, and water circulating system. Interior metal surfaces of the water system shall be preserved with P-3 or P-21.

5.1.1.8 Gages and control panels. Mounted gages and control panels shall be cushioned to prevent breakage.

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5.1.1.9 Cleaning gun. The cleaning gun and discharge hose shall be secured on the provided appurtenance in a manner to prevent damage and movement during shipment.

5.1.1.10 Hose. The water hose shall be coiled to a minimum safe diameter and secured.

5.1.1.11 Rubber tires. Tires shall be inflated to 10 pounds above the specified operating pressure.

5.1.1.12 Consolidation. If provided, toolboxes shall be utilized for the consolidation of loose components, tools, and publications. The remainder of the components that will not fit in the toolboxes shall be consolidated, along with disassembled components, in boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Boxes shall be secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

5.1.2 Level B. Level B preservation shall be the same as level A with the exception of being in accordance with method I and the following modification.

5.1.2.1 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type I, method I.

5.1.3 Commercial. The cleaner shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.1).

5.2.1 Levels A. Level A packing shall be an enclosure consisting of ends, sides, and top constructed in accordance with MIL-C-104, type I, class optional, to house the cleaner. The housing shall be secured to the cleaner base or to the trailer. The contents shall be anchored, blocked, braced, and cushioned in accordance with the crate specification and MIL-STD-1186.

5.2.2 Level B. Cleaners mounted on highway-type trailers shall be shipped uncrated. Consolidated packs, disassembled components, and accessories shall be positioned and secured to the cleaner in a manner that will not interfere with towing, or lifting with slings. Boxes shall be secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall not increase cubage or interfere with lifting or mobility of the equipment.

5.2.3 Commercial. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The cleaners covered by this specification are intended for bomb and mine disposal operations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Style required (see 1.2).
c. Issue of document required if other than as specified (see 2.1.1).
d. When a first article is required for inspection and approval (see 3.2, 4.4, and 6.3).
e. Whether a steam, high-pressure/hot-wash, or combination cleaner is required (see 3.6).
f. When a fully enclosed sheet-metal housing is required (see 3.7).
g. When a concentrated cleaning solution system is not required (see 3.13.4).
h. When a fire extinguisher is required (see 3.15.6).
i. When lubrication is to be other than as specified (see 3.17).
j. When electromagnetic interference control is required (see 3.18).
k. When lifting and tiedown attachments are required (see 3.19 and 4.5.4).
l. Required finish coat color (see 3.20).
m. When treatment for fungus resistance is required (see 3.21).
n. When an endurance test shall be performed on the first article (see 4.4.2).
o. Level of preservation and level of packing required (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.
3.5.1 and 4.5.3	Certificate of compliance	DI-E-2121

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete cleaner. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Definitions.

6.4.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt paved highway.

6.4.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

6.5 Cross-reference of classifications. The classification of cleaners in this document differs from the superseded specification in the following respects.

MIL-C-82033(NAVY)	MIL-C-82033A(YD)
Style 1	Style 1
Style 2	Not designated
Style 3	Style 2

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:

Navy - YD

Project No. 4940-N364

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-82033A(YD) - CLEANER, STEAM, PRESSURE JET, TRAILER-MOUNTED (FOR BOMB DISPOSAL OPERATIONS)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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